Records, Dissipation and Intervention: Conditions for a Temporal Direction

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Abstract

Records, entropy production and intervention need not point in the same direction. We bound disjoint fragments' Holevo information by the purity supplied at a product boundary, bath work and a possible system contribution. Fair pointer copying into independent partly mixed ports saturates the pointer-controlled bound: each port's readable information equals its initial purity deficit. The same controlled operations yield an exact joint ledger, a posterior for action and, for one port, a transfer of its purity into the source without a bath. Retained correlations provide another resource: a conditional bound prices new ports relative to an old reference even when their marginal purity is zero. Fresh ports suppress coherence and accumulate records along a CP-divisible source dynamics. Reusing a port can instead return correlations and restore coherence. Reset, reversible relabelling and reversal of the complete recording history have different balances. For a bank degraded by independent flips, restoration of the complete retained state removes a specified joint entropy increment. Label-controlled thermal swaps realize the correction for nonzero target error and approach its entropy price; a declared thermal chiral exhaust adds a conditional finite-rate heat bound. Time, coupling order, gates and preparation are inputs. The results explain how records support later intervention and what must be restored for that capacity to continue.

1 Introduction

1.1 The problem

The fundamental dynamical laws are invariant under time reversal, or nearly so, while records, memories and the reach of actions are not: we have records of the past and can influence only the future. Since Boltzmann the standard explanation has been a special initial condition, a low-entropy past, together with the statistical tendency of entropy to grow [6, 7, 8, 9]. The explanation leaves several questions open. What exactly must be special about the initial condition for records to form? Is the growth of records the same arrow as the growth of entropy, or as the dissipation of a system's free energy? Why can an intervention affect later records and not earlier ones, and what would it take to undo a record?

The paper answers these questions for record-forming processes in which records are well defined. We do not derive a time parameter. The dynamics is given with an external parameter and an order of interactions. What the paper derives is a set of quantitative relations among four directions that are often treated as one: the growth of records, the growth of entropy, the dissipation of free energy, and the reach of interventions. It shows when they coincide, when they separate, and what each costs.

1.2 Results

Section 3 recalls the record model of Paper E [3], in which fresh memories collide once with a pointer, and the two counts of records it supports. The results are:

- a purity law valid for every record-forming process that starts from a product of system, memories and thermal bath: the Holevo information of disjoint fragments about a pointer label is at most the initial purity of the memories, plus the work not stored in the system in units of k_BT, plus the difference between the label entropy and the initial system entropy (Theorem 4.1). The records version of a low-entropy past is a stock of purity or free energy, and the bound is attained;

- in the collision model, an exponential bound on the system's coherence in terms of the number of records and the corresponding loss of free energy, which disappears when the system starts diagonal (Proposition 5.1); record growth along a CP-divisible reduced dynamics (Proposition 5.2); and a gap between local and global entropy that grows linearly with the number of records (Proposition 5.3);

- thermal reset prices: at least k_BT times local entropy for separate fragment erasers in the pure-reset limit, a joint-reset infimum at most k_BT times label entropy, and zero net endpoint work for complete dynamical reversal (Proposition 6.1);

- partly mixed ports that attain the pointer-controlled purity bound, an exact acquisition–readout–action ledger and a conditional record budget with a retained reference (Propositions 4.3 and 4.4);

- an exact intervention order: operations on the system cannot change memories that no longer interact, and change records formed later (Proposition 7.2); repeated interaction breaks the alignment (Proposition 7.3).

- a separate price for renewing a noisy record bank: a complete cyclic correction removes $R[h(q) - h(\varepsilon)]$ nats, while a protected old record can require no renewal. A thermal chiral exhaust gives a conditional finite-rate capacity for such a bank (Propositions 8.1 and 8.2).

Section 9 collects the comparisons, including the two models of T-DOME I in which correlations return to a system without records or no correlations return at all [5]. Section 10 relates the results to the thermodynamic arrow, the past hypothesis and timeless formulations, and Section 10 places the paper in the HAFF programme.

2 Time in physical theory

In Newtonian mechanics and in the Schrödinger equation time is an external parameter, and the equations have no intrinsic arrow. In canonical quantum gravity the Wheeler–DeWitt equation $\hat{H}|\Psi\rangle = 0$ contains no time parameter [11]. The Page–Wootters construction recovers an effective time from correlations between a clock subsystem and the rest of a stationary universal state [12], but it explains an ordering rather than its asymmetry. The thermodynamic arrow supplies the asymmetry through a coarse-graining and a low-entropy boundary condition [9, 10].

The model below uses an external parameter and an externally given order of interactions. It asks how records become asymmetric relative to a boundary condition, and it states that boundary condition as a resource. It is not an input-free derivation of time.

3 Records and how to count them

3.1 The record model

The system S is a qubit with pointer projections P_x, $x \in \{0, 1\}$, and initial state $\sqrt{p}\,|0\rangle + \sqrt{1-p}\,|1\rangle$. Memory qubits $E_1, \ldots, E_n$ start in $|0\rangle$, and the k-th collides once with S through $U_k = P_0 \otimes \mathbf{1} + P_1 \otimes R_k$, where R_k is a real rotation with $\langle 0|R_k|0\rangle = c_k \in [0, 1]$. After n collisions

$$|\Psi_n\rangle = \sqrt{p}\,|0\rangle|e_0\rangle + \sqrt{1-p}\,|1\rangle|e_1\rangle, \qquad |e_x\rangle = \bigotimes_k |e_{x,k}\rangle, \qquad \langle e_{0,k}|e_{1,k}\rangle = c_k. \tag{1}$$

For a fragment F put $r_F = \prod_{k\in F} c_k$ and

$$H_p(r) = h\left(\frac{1 + \sqrt{1 - 4p(1-p)(1-r^2)}}{2}\right), \qquad h(x) = -x\ln x - (1-x)\ln(1-x).$$

Paper E shows that $\chi(X : E_F) = H_p(r_F)$, that the coherence of S is $\sqrt{p(1-p)}\,r_{[n]}$, and that $S(\rho_S) = S(\rho_{E_{[n]}}) = H_p(r_{[n]})$ [3]. Here X is the value a pointer measurement on S returns. Entropies are in nats.

This pure-port model will be compared with the partly mixed controlled-X model below. The inputs are the pointer basis, the split into system and memories, the fragment partition, the order of collisions with its external parameter, and the preparation of the memories. The collisions can be implemented by couplings $g_k(t)P_1 \otimes \sigma_y$ that vanish at the start and end of each pulse and commute with a pointer Hamiltonian $H_S = \varepsilon P_1$; with degenerate memory Hamiltonians the recording unitary has zero net endpoint work. Preparation of the ports, calibrated pulse control and later bank renewal are separate resources. T-DOME I gives the full thermodynamic account of this protocol [5].

3.2 Redundancy and protocol-relative comparisons

Definition 3.1 (Redundancy index)**.** *For a classical label X, a classical–quantum state ρ_{XE} and a declared family of disjoint fragments $\{E_k\}$, define*

$$\mathcal{R}_\delta^{\chi} = \#\big\{k : \chi(X : E_k) \geq (1-\delta)H(X)\big\}, \qquad \mathcal{R}_\delta^{\mathrm{acc}} = \#\big\{k : I_{\mathrm{acc}}(X : E_k) \geq (1-\delta)H(X)\big\}.$$

The first counts Holevo records and the second records that a permitted single measurement can read; $\mathcal{R}_\delta^{\mathrm{acc}} \le \mathcal{R}_\delta^{\chi}$, and Paper E gives closed-form fragment lengths for both in the collision model. With $p = 1/2$, $c_k = 0.8$ and $\delta = 0.1$, blocks of five memories are Holevo records and blocks of seven are readable, so after n collisions $\mathcal{R}_\delta^{\chi} = \lfloor n/5 \rfloor$ and $\mathcal{R}_\delta^{\mathrm{acc}} = \lfloor n/7 \rfloor$. High redundancy supports public readability of the label; it does not select one global outcome. Here the quoted pure-port counts permit arbitrary fragment POVMs, as in Paper E. A restricted readout menu needs its own accessible-information optimization; ideal algebra closure does not grant those measurements.

Definition 3.2 (Protocol-relative comparisons)**.** *Fix a dynamics, a fragment family and a label X, and let α, β be configurations along the resulting history. Write $\alpha \preceq_{\mathcal{R}} \beta$ if $\mathcal{R}_\delta(\rho_\alpha) \le \mathcal{R}_\delta(\rho_\beta)$, and $\alpha \preceq_{\mathrm{inc}} \beta$ if the algebra of memories that have interacted at α is contained in that at β. Each relation is a preorder on its declared domain.*

In the collision model both comparisons follow the collision order. They need not agree in general: decoherence may increase the number of records while the coherence-sensitive effects of the system, whose expectation values scale with $r_{[n]}$, become unreadable on S itself.

3.3 Temporal inputs

The construction uses temporal orientation at several points: the parameter of the dynamics, the order of collisions, the comparison of records at different parameter values, and a boundary condition of fresh memories at one end of the history. The comparisons above therefore compress an already timed process. The aim of the following sections is to identify what that boundary condition consists of, what record growth costs and implies, and which other directions follow it.

4 The purity of the past

Bath balances below use finite entropies, relative entropies and mean energy changes. An infinite reservoir requires a controlled limit of finite reservoirs preserving those balances, not subtraction of divergent global entropies.

Theorem 4.1 (Purity law for records)**.** *Let a system S, memories $E_1, \dots, E_n$ of dimensions d_k with degenerate Hamiltonians, and a bath B with Hamiltonian H_B start in the product state $\rho_S \otimes \bigotimes_k \sigma_k \otimes \tau_B$, where τ_B is the Gibbs state of H_B at inverse temperature β. Let them evolve under a time-dependent Hamiltonian whose interaction terms vanish at the initial and final times and whose bath and memory terms are then H_B and the given degenerate Hamiltonians, let W be the work done on the whole and ΔE_S the change of the energy of S. Let X be the outcome of a final measurement of a PVM on S with rank-one projections, and let $F_1, \dots, F_R$ be disjoint sets of memories. Then*

$$\sum_{j=1}^{R} \chi(X : E_{F_j}) \le \sum_{k=1}^{n} \big[\ln d_k - S(\sigma_k)\big] + \beta\,(W - \Delta E_S) + \big[H(X) - S(\rho_S)\big]. \tag{2}$$

Without a bath the middle term is absent, and the bound holds for an arbitrary joint unitary on S and the memories, with no condition on their Hamiltonians.

Proof. Let ρ be the final state, and $\rho' = \sum_x (P_x \otimes \mathbf{1})\rho(P_x \otimes \mathbf{1})$ its pinching in the pointer basis. Pinching does not decrease entropy and unitary evolution preserves it, so $S(\rho') \geq S(\rho_S) + \sum_k S(\sigma_k) + S(\tau_B)$. Since the P_x have rank one, $S(\rho') = H(X) + S(E_{[n]}B \mid X)$, where the conditional entropy refers to the classical–quantum state of X, the memories and the bath. Write G for the memories outside $\bigcup_j F_j$. Subadditivity of conditional entropy gives

$$S(E_{[n]}B \mid X) \leq \sum_j S(E_{F_j} \mid X) + S(G \mid X) + S(B \mid X) \leq \sum_j S(E_{F_j} \mid X) + \ln d_G + S(\rho_B),$$

with ρ_B the final state of the bath. Nonnegativity of $D(\rho_B \| \tau_B)$ gives $S(\rho_B) \leq S(\tau_B) + \beta \Delta E_B$. Hence

$$\sum_j S(E_{F_j} \mid X) \geq S(\rho_S) + \sum_k S(\sigma_k) - H(X) - \ln d_G - \beta \Delta E_B.$$

Since $\chi(X : E_{F_j}) = S(E_{F_j}) - S(E_{F_j} \mid X) \leq \ln d_{F_j} - S(E_{F_j} \mid X)$, summing over j gives (2) with $\beta \Delta E_B$ in the middle. The memory Hamiltonians are degenerate and the interactions vanish at the endpoints, so the first law gives $\Delta E_B = W - \Delta E_S$. Without a bath the same argument applies with B omitted. □

The bound is attained without a bath. If S starts in $|0\rangle$, a single memory starts maximally mixed, and a controlled-NOT with the memory as control and S as target is applied, then the memories have no purity, $H(X) - S(\rho_S) = \ln 2$, and $\chi(X : E_1) = \ln 2$. In that example the label is created by copying the memory's randomness into a pure system, which is why the system can contribute one record's worth. For a bath-free pointer-controlled unitary, $U = \sum_x P_x \otimes V_x$, the system term is absent: the conditional states of the memories are unitary images of their initial state, and the argument of the proof gives $\sum_j \chi(X : E_{F_j}) \leq \sum_k [\ln d_k - S(\sigma_k)]$. If in addition each V_x is a product over memories, as in the collision model, each fragment is bounded separately by its own purity, $\chi(X : E_F) \leq \sum_{k \in F} [\ln d_k - S(\sigma_k)]$ (Paper E's purity-deficit bound). Without the product form the per-fragment bound can fail: with E_1 maximally mixed, E_2 pure, $p = \frac{1}{2}$ and V_1 the swap of E_1 and E_2, the fragment $\{E_1\}$ holds 0.216 nats about X.

Corollary 4.2 (Purity budget of a record history)**.** *If R disjoint fragments are Holevo records at level δ, then*

$$R(1-\delta)H(X) \leq \beta \sum_k \big[F(\sigma_k) - F(\mathbf{1}/d_k)\big] + \beta\,(W - \Delta E_S) + H(X) - S(\rho_S),$$

where F is the nonequilibrium free energy at temperature T. In particular, memories that start in thermal equilibrium can hold R records only if the work not stored in the system satisfies $\beta(W - \Delta E_S) \geq R(1-\delta)H(X) - [H(X) - S(\rho_S)]$. In a pointer-controlled process without a bath, $R(1-\delta)H(X) \leq \sum_k [\ln d_k - S(\sigma_k)]$.

Proof. Each record contributes at least $(1-\delta)H(X)$ to the left side of (2). For a degenerate Hamiltonian the thermal state of a d-dimensional memory is $\mathbf{1}/d$, and $\beta[F(\sigma) - F(\mathbf{1}/d)] = D(\sigma \| \mathbf{1}/d) = \ln d - S(\sigma)$. The pointer-controlled statement is the summed bound stated after Theorem 4.1. □

The acquisition boundary is thus a quantitative resource, not a requirement that every memory be pure. Stable old records can also serve as standards across time, provided their couplings and orientation connect the old and present instruments [15]. The bound counts simultaneously present information in declared disjoint fragments, not all records that could ever be made, read, erased and remade. With product maximally mixed ports, no bath and pointer-controlled recording, it gives zero; an old correlation can change that boundary, as the conditional bound below shows. The theorem specifies what the acquisition history must supply. It does not explain the primitive origin of that stock.

4.1 Partly mixed ports: acquisition, reading and action

Let a fair classical pointer X be held in S, initially $I/2$. Each of R fresh ports starts independently in $\eta_\alpha = (I+\alpha Z)/2$, with $0 \le \alpha \le 1$. Put $\varepsilon = (1-\alpha)/2$ and $i_\alpha = \ln 2 - h(\varepsilon)$. Apply controlled X from S to each port, using the same physical mixed-port coupling as Paper E. The resulting state is diagonal:

$$P(x, \boldsymbol{y}) = \tfrac{1}{2} \prod_{j=1}^{R} (1-\varepsilon)^{\mathbf{1}_{y_j = x}} \varepsilon^{\mathbf{1}_{y_j \neq x}}. \tag{3}$$

An actual Z reading obtains Y_j. The stated calibration fixes which port value labels which pointer; a common unknown reversal would not be removed by collecting more ports.

Proposition 4.3 (A saturated acquisition and action ledger)**.** *Every port has Holevo and accessible information i_α, equal to its initial purity deficit. Write $I_R = I(X : \boldsymbol{Y})$. Then*

$$\begin{aligned} S(SE) &= \ln 2 + Rh(\varepsilon), & S(E \mid X) &= Rh(\varepsilon), \\ S(E_j) &= \ln 2, & S(E) &= Rh(\varepsilon) + I_R, \\ \sum_j S(E_j) - S(E) &= Ri_\alpha - I_R, & H(X \mid \boldsymbol{Y}) &= \ln 2 - I_R. \end{aligned} \tag{4}$$

For $R = 1$, feedback controlled by the actual port, $D = X \oplus Y$, leaves the source in η_α and the port in $I/2$, independently. For degenerate source and port Hamiltonians, the source free energy increases by $k_B T i_\alpha$, at zero net endpoint work for the ideal recording and feedback unitaries.

Proof. Conditional ports have entropy $h(\varepsilon)$, while their fair-label averages are $I/2$. Their states commute, so Z reading attains Holevo information. Thus $\sum_j \chi = Ri_\alpha$ saturates the pointer-controlled bound for any number of independent ports, including partly mixed ones. Conditional product structure gives $S(SE)$ and $S(E \mid X)$; the definition of I_R gives $S(E)$. For one port, $Y = X \oplus N$, where N is independent with $\Pr(N = 1) = \varepsilon$. Fair X makes Y independent of N; feedback gives $D = N$. Both gates are basis permutations and preserve the complete joint entropy. The source entropy falls from $\ln 2$ to $h(\varepsilon)$, giving the free-energy increment. □

The posterior is available from the actual record, not just from the algebra it generates. For $k = \sum_j y_j$,

$$\Pr(X = 1 \mid \boldsymbol{y}) = \frac{(1-\varepsilon)^k \varepsilon^{R-k}}{(1-\varepsilon)^k \varepsilon^{R-k} + (1-\varepsilon)^{R-k} \varepsilon^k}, \tag{5}$$

with the deterministic limits interpreted on their support. A reversible gate controlled by $\boldsymbol{y}$ can send X to $X \oplus g(\boldsymbol{y})$. Majority is the minimum-error choice here; ties may be resolved by a fixed label. All original ports remain. Computing and retaining an extra classical copy or a decision register would require its own physical stock, so the ideal implementation uses the ports themselves as controls.

The one-port result locates the resource after action: initial memory purity has moved into the source, not disappeared into an unpriced bath. It supplies free energy for a subsequent task, not work already delivered to a work store. T-DOME I implements such a task and accounts for its reset. Neither the source nor the bank has completed a thermodynamic cycle. Paper C's blank-body transfer is another use of finite stock, with different dynamics; its resource account must be retained when the two mechanisms are composed.

4.2 Old references change the acquisition boundary

Proposition 4.4 (Conditional budget with a retained reference)**.** *Let X be classical, K a retained reference, and $E = E_1 \otimes \cdots \otimes E_R$, with $\dim E_j = d_j$. The initial state may correlate all of them. Recording applies $\sum_x |x\rangle\langle x| \otimes I_K \otimes V_x$, with V_x acting on E only. Then*

$$\sum_j I(X : E_j \mid K)_{\text{out}} \leq \sum_j \ln d_j - S(E \mid XK)_{\text{in}}. \tag{6}$$

For independent fresh product ports the right side is their summed purity deficit. This bounds final conditional information, not only its increment during recording.

Proof. The controlled unitary preserves $S(E \mid XK)$. Strong subadditivity gives $\sum_j S(E_j \mid XK)_{\text{out}} \geq S(E \mid XK)_{\text{out}}$. Also $S(E_j \mid K) \leq S(E_j) \leq \ln d_j$. Subtracting these inequalities proves the claim. □

Take independent fair bits X, L, an old reference $K = L$ and a new port $E = L$. Each port is marginally maximally mixed. Controlled X writes $Y = L \oplus X$ into E; E alone carries no information about X, but reading K, E recovers it exactly. Here $S(E \mid XK)_{\text{in}} = 0$ and $I(X : E \mid K)_{\text{out}} = \ln 2$, attaining Eq. (6). The usable stock is the old correlation and permission to read the reference jointly, not a hidden pure marginal. Creating and retaining the two correlated copies of L belongs to the earlier history's account.

For classical readable K, the conditional information is the Holevo quantity averaged over its records, and in this commuting example it is actually attained. A quantum reference need not admit that interpretation: the entropy inequality remains valid, but its conditional mutual information is not by itself a measurement permission or accessible information. This distinction connects acquisition to T-DOME I's retained side information without identifying unlike tasks.

5 Records and dissipation

Proposition 5.1 (Records bound the coherence that survives)**.** *In the collision model with $0 < p < 1$, let $0 < \delta < 1$, let q_δ satisfy $h(q_\delta) = (1-\delta)h(p)$ with $q_\delta \leq 1/2$, and put*

$$a_\chi = \frac{(1-2q_\delta)^2 - (2p-1)^2}{4p(1-p)} \in (0,1).$$

If R disjoint fragments, of any sizes and any collision overlaps c_k, are Holevo records, then

$$r_{[n]}^2 \le a_\chi^R, \qquad |(\rho_S)_{01}| \le \sqrt{p(1-p)}\, a_\chi^{R/2}.$$

If the pointer Hamiltonian is diagonal in the pointer basis, the system's nonequilibrium free energy falls by

$$F_S(0) - F_S(n) = k_B T\, H_p(r_{[n]}) \ge k_B T\, H_p\big(a_\chi^{R/2}\big).$$

If instead S starts in $pP_0 + (1-p)P_1$, every record quantity is unchanged and $F_S(n) = F_S(0)$.

Proof. A fragment is a Holevo record exactly when $r_F^2 \le a_\chi$ (Paper E's Holevo fragment criterion). The overlaps multiply over disjoint fragments, and the memories outside them contribute factors at most 1, which gives the first two bounds. The collisions preserve the pointer populations, so the energy of S is unchanged and $F_S(0) - F_S(n) = k_B T[S(\rho_S(n)) - S(\rho_S(0))] = k_B T\, H_p(r_{[n]})$; H_p decreases in r. For the diagonal initial state the conditional fragment states are the same, so all $\chi(X : E_F)$ are unchanged, and ρ_S does not change at all. □

Records and dissipation in the system therefore coincide only when the system has coherence to lose. As R grows, $H_p(a_\chi^{R/2}) \to h(p)$ and the system loses all the free energy carried by its coherence. A system that is already diagonal acquires the same records at no cost to itself. In the pure-port model, the marginal purity of written memories is transferred into correlations. The complete unitary preserves joint entropy; the mixed-port ledger makes that transfer explicit.

Proposition 5.2 (Records along a CP-divisible dynamics)**.** *The reduced dynamics of S after n collisions is $\mathcal{D}_n \circ \cdots \circ \mathcal{D}_1$, where $\mathcal{D}_k$ keeps the populations and multiplies the coherence by c_k. Every intermediate map $\mathcal{D}_m \circ \cdots \circ \mathcal{D}_{l+1}$ is completely positive and trace preserving, and for equal overlaps $c \in (0,1)$ applied at intervals Δt the family embeds in the dephasing semigroup generated by $\mathcal{L}(\rho) = \frac{\gamma}{2}(Z\rho Z - \rho)$ with $\gamma = -\ln c/\Delta t$. Along this dynamics $\mathcal{R}_\delta^\chi$ and $\mathcal{R}_\delta^{\mathrm{acc}}$ grow without bound in n.*

Proof. Each $\mathcal{D}_k$ is a dephasing channel, and compositions of channels are channels. The semigroup multiplies the coherence by $e^{-\gamma t}$, which equals c^m at $t = m\Delta t$. The record counts are $\lfloor n/m_\chi \rfloor$ and $\lfloor n/m_{\mathrm{op}} \rfloor$ for equal overlaps. Unbounded record counts refer to a family with growing fresh-port supply, not an unbounded yield from a fixed finite bank. □

Record growth thus does not require memory effects in the reduced dynamics of the system. The memories, once written, have a mixture-of-product marginal correlated through X and are independent conditional on a pointer reading; a Markovian system dynamics says nothing about whether the environment's outputs are correlated. The converse separation also holds. T-DOME I gives an autonomous two-qubit model in which correlation built between a system and an environment qubit returns to the system at a recurrence time while the environment holds no information at all about the pointer, and it shows that the pure-dephasing spin–boson model with a Lorentz–Drude spectral density has no backflow at any temperature [5]. Backflow and records are logically independent.

Proposition 5.3 (Local and global entropy)**.** *For disjoint fragments $F_1, \dots, F_R$ of the collision model,*

$$\sum_{j=1}^{R} S(\rho_{F_j}) = \sum_j \chi(X : E_{F_j}), \qquad S\big(\rho_{\cup_j F_j}\big) \le h(p).$$

If the F_j are Holevo records, the total correlation among them, $\sum_j S(\rho_{F_j}) - S(\rho_{\cup_j F_j})$, is at least $[R(1-\delta)-1]h(p)$.

Proof. The conditional states of every fragment are pure, so $S(\rho_{F_j}) = \chi(X : E_{F_j})$. The union of the fragments has conditional states that are pure as well, so $S(\rho_{\cup F_j}) = H_p(r_{\cup F_j}) \le H_p(0) = h(p)$. Subtracting gives the last claim. □

With $p = 1/2$, $c = 0.8$ and $n = 60$ memories in twelve blocks of five, the local entropy is 7.661 nats while the entropy of all sixty memories is 0.693147 nats, equal to $\ln 2$ to nine digits. A reader with access to single fragments sees entropy grow with every record. The global entropy of the environment saturates at the entropy of the label. Which of the two grows without bound is fixed by the access structure, not by the dynamics.

6 Undoing records

Proposition 6.1 (Local and joint reset)**.** *Let the pure-port model's memories have degenerate Hamiltonians. Each eraser couples the memories it controls to its own independent Gibbs bath at inverse temperature β, with interactions off at endpoints. A reset with final memory entropy tending to zero has:*

1. *for separate fragment erasers,*

$$\liminf W^{\text{local}} \ge k_B T \sum_j S(\rho_{F_j}) = k_B T \sum_j \chi(X : E_{F_j}) \ge R(1-\delta) k_B T h(p)$$

   when those fragments are Holevo records;

2. *for one eraser controlling all memories but not S, reset work infimum $k_B T S(\rho_{E_{[n]}}) = k_B T H_p(r_{[n]})$, approached with vanishing reset error and increasingly large, engineered thermal reservoirs;*

3. *for an agent controlling S and every memory, the inverse $U_n^{-1} \cdots U_1^{-1}$ restores the initial product state and source coherence at zero net endpoint work in the ideal recording model.*

Proof. For one controlled memory M and its initially independent bath, unitarity and subadditivity give $\Delta S_B \ge S(\rho_M) - S(\rho'_M)$. The Gibbs identity gives $\beta W = \beta Q = \Delta S_B + D(\rho'_B \| \tau_B)$, since the degenerate memory energy is unchanged. Hence $\beta W \ge S(\rho_M) - S(\rho'_M)$. Apply this separately, or to the entire bank. Separate baths cannot exploit correlations between fragments; their marginal inputs fix these entropy decreases. The pure-port bank marginal has rank at most two. A unitary acting only on the degenerate bank maps its two eigenvectors to a logical qubit tensored with blank remaining bits. The logical qubit has eigenvalues $(1-z, z)$, $0 \le z \le 1/2$, and entropy $h(z) = H_p(r_{[n]})$. For $z > 0$, thermal swaps through decreasing positive error probabilities to $\epsilon > 0$ approach

$k_B T[h(z)-h(\epsilon)]$, as in Proposition 8.2. Taking $\epsilon \to 0$ gives the joint reset infimum; for $z = 0$ the bank unitary suffices. The same thermal-swap method applies more generally [14]. The inverse recording unitaries have the same unchanged endpoint energy as the forward gates. □

Exact purification of a mixed memory is not attainable with a finite-dimensional, full-rank Gibbs bath at finite energies and temperature. Its input joint rank is $\mathrm{rank}(\rho_M)d_B > d_B$, whereas a pure final marginal would force the joint state to have rank at most d_B. Unitary evolution preserves rank. Thus the work infimum is not an exact finite-bath reset guarantee; finite reset error uses the entropy decrease in the proof, with nonnegative bath relative-entropy and correlation terms.

For $p = 1/2, c = 0.8, n = 60$, the corresponding asymptotic reset bounds are $19.505\, k_B T$ for sixty separate ports, $7.661\, k_B T$ for twelve five-port fragments, $5.368\, k_B T$ for eight seven-port readable fragments, and $0.693\, k_B T$ for the joint bank. The local price reflects the access structure, not a larger amount of distinct label content.

Lemma 6.2 (Reset with an old reference)**.** *Let M, K have zero endpoint Hamiltonians and the bath be thermal and independent of MK. Allow a joint unitary on MKB that leaves the marginal of K unchanged. Then*

$$\beta W = \beta Q \geq S(M \mid K)_{\mathrm{in}} - S(M').$$

In the pure-reset limit the lower bound is $S(M \mid K)_{\mathrm{in}}$.

Proof. The decrease of the retained joint entropy is $S(MK)_{\mathrm{in}} - S(M'K')_{\mathrm{out}}$. Since $S(M'K') \leq S(M') + S(K') = S(M') + S(K)_{\mathrm{in}}$, it is at least $S(M \mid K)_{\mathrm{in}} - S(M')$. The independent-bath argument above applies to the joint retained state. □

The target here preserves the declared reference marginal. If other reference correlations must also be restored, they belong to the retained target and its ledger. Quantum conditional entropy is not a universal attained work price without an allowed protocol. For two classical copies $M = K = L$ of a fair bit, controlled XOR from K resets M exactly at zero endpoint work and leaves K unchanged. Its conditional entropy is zero, although $S(M) = \ln 2$. Permission to use the reference changes the reset task.

Changing a record is not resetting it. A unitary bit flip reverses its labels at zero endpoint work for a degenerate memory and retains all information. Swapping a record into an independent maximally mixed port makes the old memory uninformative at zero endpoint work, but transfers the record into the other port instead of destroying the joint history. Local thermal reset, such transfer, error correction and inversion of the complete source–bank unitary cannot be assigned one erasure bill. Exact history reversal is allowed when all required gates and systems are accessible. Neither its probability nor a suppression of reversed histories follows without an ensemble and measure [16].

7 Intervention and the agent's time order

Definition 7.1 (Protocol-relative causal-influence diagnostic)**.** *Fix a record-forming protocol with external parameter t, labels X_A, X_B, and a family of interventions on X_A. Write $X_A \rightsquigarrow_{\mathsf{P}} X_B$ when (i) both labels have records in the declared fragments, (ii) the X_A records form no later than the X_B records according to t, and (iii) interventions on X_A, with the rest of the protocol held fixed, change the distribution of X_B or its records.*

The diagnostic is not asserted to be transitive, antisymmetric or independent of the intervention family. Redundancy makes an influence readable across fragments; the intervention structure carries the causal content. The next two propositions compute the diagnostic in the collision model.

Proposition 7.2 (Interventions reach only later records)**.** *In the collision model let an arbitrary channel $\mathcal{T}$ act on S between the j-th and $(j+1)$-th collisions.*

1. *The final reduced state of $E_1, \dots, E_j$ is the same as without $\mathcal{T}$.*
2. *If $\mathcal{T}$ is the pointer flip $\rho \mapsto \sigma_x \rho \sigma_x$, the final state is $\sum_x \sqrt{p_x}\,|1-x\rangle|e_x^{\le j}\rangle|e_{1-x}^{>j}\rangle$: the earlier memories record the value x and the later ones record $1-x$.*
3. *If the Kraus operators of $\mathcal{T}$ are diagonal in the pointer basis, the joint distribution of the pointer value and all records is unchanged.*

Proof. (1) After the j-th collision, $\mathcal{T}$ and the later collisions act on $S \otimes E_{j+1} \otimes \cdots \otimes E_n$, and the partial trace over these systems is invariant under any operation on them. (2) σ_x maps the branch $\sqrt{p_x}|x\rangle|e_x^{\le j}\rangle$ to $\sqrt{p_x}|1-x\rangle|e_x^{\le j}\rangle$, and the later collisions then write the value $1-x$. (3) Diagonal Kraus operators commute with the pointer-controlled collisions and with the pointer measurement, and they leave the populations unchanged. □

For a flip after the second of four collisions with overlaps $(0.3, 0.5, 0.2, 0.4)$ and $p = 1/2$, the reduced states of the first two memories and of the last two are each unchanged, while the joint state of the first and third memories changes by 0.437 in its largest matrix element: the flip is visible only in the relation between earlier and later records. In this fresh-port protocol, interventions on the source can alter later records, while detached earlier marginals remain fixed. The flip example makes the change visible in joint relations even when single-fragment marginals do not change.

Proposition 7.3 (Repeated interaction lets a later intervention rewrite an earlier record)**.** *Let a memory collide with S twice through the same rotation with $c = \cos(\pi/4)$, and let $p = 1/2$. After the first collision the memory holds $H_{1/2}(2^{-1/2}) = 0.4165$ nats about the initial pointer value. If nothing happens between the collisions, the second collision makes the record perfect, $\ln 2$ nats. If S is flipped between the collisions, the second collision leaves the memory with no information about the initial pointer value.*

Proof. Write R for the rotation by $\pi/4$, so $R^2|0\rangle = |1\rangle$. Without the flip, the memory ends in $|0\rangle$ if $x = 0$ and in $R^2|0\rangle = |1\rangle$ if $x = 1$. With the flip, it is in $|0\rangle$ after the first collision if $x = 0$ and is then rotated once, ending in $R|0\rangle$; if $x = 1$ it is in $R|0\rangle$ after the first collision and is not rotated again. Both branches end in $R|0\rangle$. □

For the coherent source $|+\rangle$, the same two-qubit protocol also gives the entropy and population ledger:

$$|\Psi_1\rangle = (|0,0\rangle + |1\rangle R|0\rangle)/\sqrt{2}, \quad |\Psi_{2,\text{noflip}}\rangle = (|00\rangle + |11\rangle)/\sqrt{2}, \quad |\Psi_{2,\text{flip}}\rangle = |+\rangle \otimes R|0\rangle.$$

Source populations stay $1/2, 1/2$, and joint entropy stays zero. Put $a = h[(1+2^{-1/2})/2] = 0.4164955$. At first contact, $S(S) = S(E) = a$ and $I(S:E) = 2a$. The no-flip continuation gives $\ln 2, \ln 2, 2\ln 2$. The flip continuation gives $0, 0, 0$: source coherence and both marginal purities return while the earlier pointer information disappears. For a pointer-diagonal source Hamiltonian, energy stays fixed; source free energy first falls by $k_B T a$

and then recovers it on the flip branch. In fact $U(X_S \otimes I)U = X_S \otimes R$, so the echo decouples every source input. Reading or discarding the port after the first contact would change this coherent protocol. The source trace distance of the inputs $|+\rangle, |-\rangle$ falls to $2^{-1/2}$ at first contact and returns to 1 after the echo: this is distinguishability backflow, not merely an entropy change.

The alignment of the intervention order with the record order thus rests on a specific feature of the dynamics: memories that have recorded do not interact with the source again. Under repeated interaction, whether an early memory ends as a perfect record or as no record can be decided by an intervention made after it first recorded.

An agent can therefore date an intervention by its effect on records within the stated one-contact architecture. It cannot change a detached port by acting only on the source. Acting directly on that port can relabel, transfer or reset it, with the different balances above; reconnecting it to the source can also restore coherence. The alignment comes from the actual interaction menu, fresh stock and external order, not from a universal prohibition on changing the past.

8 Renewing records instead of making new ones

A written distinction and a renewable distinction are different resources. The purity law prices acquisition from its declared product boundary. It does not make an old record disappear when fresh blank memories run out. For example, a copied Z label remains unchanged under Z dephasing, although each fair-label record has zero marginal purity. Maintaining it costs no entropy removal in that noise model. Independent bit flips give a different, quantitative renewal problem.

Proposition 8.1 (The entropy price of a cyclic record bank)**.** *Let a classical label X be retained unchanged, and let R degenerate binary registers C_i be conditionally independent copies of X with error $\varepsilon \in [0, 1/2)$. Between corrections each undergoes an independent flip of probability $p \in (0, 1/2]$, whose noise port is inaccessible. The controller starts independent of XC and is unchanged by this noise. A correction returns the complete joint state of X, the bank and its controller to the pre-noise state. Its bath starts thermal and independent, and its interactions vanish at the endpoints. Put*

$$q = p + (1 - 2p)\varepsilon, \qquad d_{p,\varepsilon} = h(q) - h(\varepsilon) > 0. \tag{7}$$

The entropy removed from the retained state is exactly $Rd_{p,\varepsilon}$ per correction, and its bath heat obeys $\beta Q \geq Rd_{p,\varepsilon}$.

If corrections occur at frequency ν and the sole exhaust has the thermal, zero-charge, extensive-current properties of HAFF-D's chiral exhaust [2], the mean heat current obeys

$$\overline{\dot{Q}} \geq k_B T \nu R d_{p,\varepsilon} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} (\nu R d_{p,\varepsilon})^2. \tag{8}$$

Thus an exhaust-power budget $\overline{\dot{Q}} \leq J$ permits at most

$$R \leq \frac{\sqrt{(k_B T)^2 + 4\alpha J} - k_B T}{2\alpha \nu d_{p,\varepsilon}}, \qquad \alpha = \frac{3\hbar}{\pi c_{\mathrm{ex}}}. \tag{9}$$

Here c_{ex} is dimensionless chiral capacity, and $\nu R d_{p,\varepsilon}$ is the dimensionless entropy removed per unit time, in nats per time. The current bound is a long-window extensive average,

not an individual pulse identity. No inference from vanishing budget to vanishing records follows when $d_{p,\varepsilon} = 0$.

Proof. Before noise $S(XC) = H(X) + Rh(\varepsilon)$, and after it $S(XC) = H(X) + Rh(q)$. Cyclicity of the entire retained state prevents a controller or a correlation with the label from absorbing that entropy without charge. For its independent Gibbs bath B, a joint unitary and subadditivity give $\Delta S_B \geq Rd_{p,\varepsilon}$. The Gibbs identity $\beta Q = \Delta S_B + D(\rho'_B \| \tau_B)$ proves the heat bound. The actual net extensive entropy current is at least $\nu R d_{p,\varepsilon}$. Substitution into HAFF-D's increasing finite-rate bound gives Eq. (8); inversion gives Eq. (9). □

Proposition 8.2 (Thermal correction with an accessible label)**.** *Under the renewal proposition's conditional-iid error law, assume* $0 < \varepsilon < q \leq 1/2$*, and allow retained* X *to control reversible XOR gates on each bank register. Supply an independent Gibbs bath qubit with gap* $\Delta_\varepsilon = k_B T \ln[(1-\varepsilon)/\varepsilon]$ *for each register. XOR by* X*, swap with that bath qubit, and XOR by* X *again restore the complete retained* X, C*, controller target. Per register the bath heat and mean driven work are*

$$Q = W = k_B T (q - \varepsilon) \ln[(1-\varepsilon)/\varepsilon]. \tag{10}$$

With $q = t_0 > t_1 > \cdots > t_m = \varepsilon$*, successive swaps with independent Gibbs qubits of gaps* $\Delta_i = k_B T \ln[(1-t_i)/t_i]$ *give*

$$\beta Q = \beta W = \sum_i (t_{i-1} - t_i) \ln[(1-t_i)/t_i] \longrightarrow h(q) - h(\varepsilon) \tag{11}$$

as the partition mesh tends to zero. All stages have finite dimensions and finite gaps for a fixed positive target error. The finite-step excess is exactly

$$\beta Q - [h(q) - h(\varepsilon)] = \sum_i D(\mathrm{Bern}(t_{i-1}) \| \mathrm{Bern}(t_i))$$

per register; the spent baths end independent of the retained target.

Proof. The first XOR converts $C_i = X \oplus N_i$ into the independent error bit N_i of probability q, without changing X. The thermal swap replaces it by an independent error bit of probability ε; the second XOR restores the intended conditional law. The bath output has probability q, so its energy gain is $\Delta_\varepsilon(q - \varepsilon)$. Controllers remain untouched, and conditional independence is restored. For several swaps, the output of bath qubit i has probability t_{i-1}. Summing its energy gain gives Eq. (11). The right-endpoint Riemann sums tend to $\int_\varepsilon^q \ln[(1-t)/t]\, dt = h(q) - h(\varepsilon)$. The degenerate retained registers have no endpoint energy change, so the first law gives $W = Q$. Each swap's heat equals its error-bit entropy decrease plus $D(\mathrm{Bern}(t_{i-1}) \| \mathrm{Bern}(t_i))$. Telescoping gives the stated finite-step excess. All initial error and bath bits are independent after the first XOR, and the swaps merely permute them; hence the spent baths are independent of the final target. □

This is a label-preserving use of the thermal swap construction of Ref. [14]. These swaps are driven gates between degenerate bank bits and nondegenerate bath qubits, not spontaneous energy-conserving collisions. The label's control permission, thermal preparations, clock and switching belong to the implementation. Spent bath qubits retain the exported entropy and are not reset by the construction. It realizes cyclicity of

the declared retained bank, not an autonomous battery-complete apparatus or an indefinitely recycled bath. The vanishing-error limit $\varepsilon \to 0$ also requires growing gaps or an appropriate limiting reservoir. This thermal-qubit implementation does not establish the separate sole-chiral-exhaust conditions or sharpness of its finite-rate bound.

For $p = .12$ and $\varepsilon = .05$, one correction removes .2378232484 nat per register. Eight correlated copies of one fair label require 1.9025859874 nats of renewal, despite carrying only one bit of distinct label content. Their independent degradation, not the number of distinct labels, sets this maintenance bill. The cyclic target and the independent noise are substantive inputs. Jointly correlated errors, an accessible noise record, a noncyclic pure controller or a protected bank can change the cost. The heat-current claim additionally needs the actual exhaust assumptions; it is not a Landauer bound on every control pulse.

This gives the record side of a maintained observer. T-DOME III and Brain Anchor let retained history choose which access to repair [17]; Eq. (9) prices one implementation of the retained bank on which such a choice relies. Foundation's exact fixed algebras describe passive protection, not a requirement to export positive entropy indefinitely. Under HAFF-D's further matched horizon-flux conditions, the bank's removal rate can enter the conditional horizon-entropy bound. Those conditions connect a real renewal process to gravity; record existence alone does not.

9 Which arrows coincide

Table 1 collects the directions studied above in four specified settings: the collision model with a coherent or a diagonal initial system state, the collision model with one repeated collision, and the autonomous two-qubit model of T-DOME I, in which a system qubit and an environment qubit build and return correlation without any pointer record [5].

Direction	collision, coherent S	collision, diagonal S	one repeated collision	two-qubit recurrence
Record count grows	yes	yes	can fall	no records
Local entropy of fragments grows	yes	yes	can fall	constant E
Marginal memory purity	falls	falls	falls, then returns	no fresh bank
System free energy falls	yes	no	falls, then returns	falls, then returns
Interventions reach only later records	yes	yes	no	no records
Local thermal reset bound $\propto R$	yes	yes	no final record	—
Correlation flows back to S	no	no	yes, with flip	yes

Table 1: Specified protocols, not universal arrows. The repeated-contact column uses the coherent, unmeasured flip continuation of Proposition 7.3. Reset prices refer to blank targets, not arbitrary changes of record content.

With pure fresh ports interacting once, record accumulation, local entropy growth and decreasing marginal port purity align; detached marginals also fix which source

interventions can still reach them. Local thermal reset prices that bank relative to a blank target. These relations do not make every record update an entropy-producing operation. The mixed-port ledger shows that acquisition and action redistribute finite stock, while the echo reverses the local entropy and correlation changes. Source free-energy loss depends on initial coherence. Reduced-system backflow, record acquisition and bank renewal remain separate questions.

10 Relations and interfaces

The purity law gives the record boundary of the past hypothesis a quantitative form, while separating record accumulation from a source's loss of free energy [9, 10]. Mixed-environment redundancy [13] is one instance; the theorem also prices stock drawn from work and a bath. Page–Wootters correlations can supply ordering within a stationary state [12]. A direction of recording additionally needs the acquisition boundary and actual contact architecture specified here.

Complete-access reversal is compatible with retrocausal programmes [16]; restricted thermal reset is a different task. The causal-influence diagnostic computes supplied interventions in a physical recording process. It does not turn them into a derived causal partial order.

Paper E supplies the recording instruments [3]. The present posterior, intervention order and reference-dependent budgets are the record-side interfaces to T-DOME I's thermodynamic agent [5]; Paper C implements a separate maintained sensing body [1]. Paper D's heat and horizon bounds apply only under their matched source conditions [2]. The purity law supplies a physical input for Paper G's question of how selection permissions form [4]; it does not derive those permissions, clocks or a geometric algebra.

11 Conclusion

The product-boundary purity law bounds simultaneously present fragment information. Independent partly mixed ports attain the pointer-controlled bound, and their actual readout supplies a posterior for action. The complete ledger tracks where purity and correlations go; a one-port feedback transfers source free energy without already extracting work. A retained reference can supply conditional stock even when a new port is marginally maximally mixed.

Fresh one-contact ports align record acquisition, loss of coherence and restricted source intervention in a particular architecture. Recontact can instead erase the earlier record and recover source coherence. Reversible relabelling, transfer, thermal reset and exact history reversal are distinct operations. Their prices depend on the target, retained references and accessible gates, not on the word *undoing*.

A noisy record bank poses a further task. Independent errors increase its complete retained entropy; label-controlled thermal swaps restore that target for positive error and approach the entropy-removal bound. A specified chiral exhaust additionally limits the correction rate. This gives a nonempty maintenance mechanism without claiming a renewable autonomous universe or the origin of its clocks and couplings. A broader theory must explain how fragment and interaction structures form, how correlated noise changes renewal, and how actual entropy export participates in source and geometric response.

A Joint-state checks

The mixed-port check constructs the complete diagonal density operator and applies the recording CNOTs, not separately chosen conditional laws. For $\alpha = 0, .2, .8, 1$ and $R = 1, 2, 3, 5$, it verifies Eq. (4), the actual Z-record posterior and one-port feedback. The maximally mixed keyed example is an explicit three-bit permutation. Forty full-density examples with a quantum reference and two initially correlated ports test the conditional bound; the entropy proof establishes it generally.

The coherent echo check applies both collisions and the intermediate flip to the same pure state, verifying $UX_SU = X_S \otimes R$, all marginal and joint entropies, and the recovery of coherence. For renewal, an eight-input XOR–SWAP–XOR permutation verifies the full label–bank target. Thermal chains with $1, 4, 16, 64, 256, 1024$ stages verify the heat and its relative-entropy excess. At $p = .12, \varepsilon = .05$, one stage costs $0.31799941\, k_B T$ per register; 1024 stages cost $0.23789030\, k_B T$, approaching $0.23782325\, k_B T$. These checks certify the finite-dimensional specified circuits, not the physical chiral exhaust.

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